Alamein International University
Computer Science & Engineering

VisionPath

AI-Powered Navigation & Conversational Assistant for the Visually Impaired

A project submitted in partial fulfillment of the requirements of the BSc. degree

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Dr. Hassan provided consistent technical direction during weekly review sessions, challenged our architectural assumptions, and helped us navigate the complex trade-offs involved in building a safety-critical real-time system.

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The department's emphasis on interdisciplinary work — combining AI, security, mobile development, and human-centred design — directly shaped the scope and rigour of this project.

1.3 Community Inspiration

Special appreciation is owed to the visually impaired community, whose lived experiences are at the centre of every design decision within VisionPath. The shared insights and everyday realities of visually impaired students and the broader Egyptian blind community illuminated the critical real-world pain-points — particularly the failure of GPS indoors, the danger of high-latency feedback during navigation, and the privacy concerns associated with volunteer-based assistance platforms — that this project directly addresses.

UNDERTAKING

This is to declare that the project entitled "VisionPath: AI-Powered Navigation & Conversational Assistant" is an original work completed by the undersigned, in partial fulfillment of the requirements for the BSc. degree at the Faculty of Computer Science & Engineering (CSE), Alamein International University (AIU).

All research, analysis, design, architecture decisions, security implementations, and system development presented in this report have been accomplished by the undersigned team members.

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ABSTRACT

VisionPath is an AI-powered mobile application designed to empower visually impaired individuals through real-time indoor navigation, object detection, and context-aware conversational assistance.

The system integrates multiple cutting-edge technologies: YOLOv8 for on-device edge computer vision, OpenAI Whisper for automatic speech recognition, Google Gemini / LLaMA as the large language model (LLM) brain, and a text-to-speech engine delivering natural spatial audio feedback.

The application enables blind users to navigate complex indoor environments — such as university campuses, hospitals, and shopping centres — without reliance on pre-mapped Wi-Fi beacons, ask natural language questions about their surroundings (e.g., "Is the lecture hall door open?" or "Did I drop my keys near this desk?"), and locate small objects like keys or phones using 3D distance estimation derived from YOLO bounding boxes and a single monocular camera feed.

Security is a first-class architectural concern: all data in transit is protected with TLS 1.3 and certificate pinning against man-in-the-middle attacks; data at rest is encrypted with AES-256; and authentication is handled through FaceID biometrics combined with Role-Based Access Control (RBAC).

VisionPath's design prioritises user autonomy, privacy, and accessibility, addressing a critical gap in currently available assistive technology solutions — particularly for the estimated 1.5 million visually impaired individuals in Egypt and the over 2.2 billion people globally affected by visual impairment (WHO, 2024).

Unlike existing solutions such as GoodMaps (which requires costly pre-mapped beacon infrastructure) or Be My Eyes (which relies on human volunteers, raising significant privacy concerns), VisionPath operates in any indoor environment without infrastructure investment and without exposing the user's private surroundings to third parties.

The project is designed as a foundation for a deployable product rather than a purely academic prototype.

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Chapter 1: Introduction & Vision

1.1 Vision and Motivation

VisionPath was born from a fundamental observation: the gap between what modern AI systems can perceive and what visually impaired individuals can independently access in their daily physical environment remains vast, dangerous, and largely unaddressed by the current assistive technology market.

The project's core mission is to close that gap through three integrated capabilities: visual intelligence that translates the physical world into semantic data in real time, conversational AI that allows natural voice-based queries about one's environment, and a security-first architecture that ensures user autonomy and privacy are never compromised in pursuit of functionality.

1.1.1 The Motivation: Ahmed's Story

Consider Ahmed, a blind university student navigating the complex indoor hallways of a large Egyptian campus daily.

Ahmed relies on a simple obstacle beeper — a device that emits an audio tone when a physical object is within proximity.

This tool keeps him from walking into the walls. But it tells him nothing about what those walls contain, where the nearest seat is in a crowded lecture hall, whether the bathroom door ahead of him is open or closed, or whether the object he just heard of fall is his phone or his keys.

Ahmed's questions reveal a fundamental semantic gap in current assistive technology: the tools available to him detect physical obstacles at a binary level, but cannot understand or describe the environment in meaningful, contextual, natural language.

Every day he must rely on asking nearby students or staff for information that a sighted person would simply observe — a constant, low-grade erosion of independence that has measurable effects on academic performance and psychological wellbeing.

VisionPath is the AI companion that Ahmed needs. When he asks, "Is the lecture hall door open?", VisionPath's YOLOv8 model identifies the door from the camera feed, estimates its state (open/closed) via bounding box analysis, and delivers the answer as natural spoken language within 200 milliseconds (Pre-Fine-Tuning).

When he asks, "Where is the nearest empty seat?", the LLM synthesises YOLO object tags, distance estimates, and spatial mapping to respond: "There is an empty chair 2 metres to your left, at 9 o'clock."

This is the vision.

1.1.2 Three Pillars of the VisionPath Mission

Visual Intelligence: Translating the physical world into real-time semantic data via Edge AI.

YOLOv8 runs directly on the mobile device at 30+ FPS, eliminating dangerous cloud round-trip latency for safety-critical object detection.

The visual intelligence layer is not a feature — it is the foundation on which every other capability rests.

Conversational AI: Providing natural voice interaction that delivers detailed situational awareness.

Rather than beep-coded alerts, VisionPath delivers full natural language responses via a large language model that combines visual context tags with the user's spoken query to generate answers that a knowledgeable, sighted companion might provide.

Autonomy & Privacy: Reducing dependence on human helpers while ensuring absolute data security.

VisionPath's multi-layer security architecture — AES-256 encryption at rest, TLS 1.3 in transit, FaceID authentication, and RBAC — ensures that the system can be trusted with the most sensitive environments: the user's home, personal documents, medical appointments, and private conversations.

1.2 Team Structure and Responsibilities

VisionPath was developed by a team of five Computer Science & Engineering students, each taking primary ownership of a distinct technical domain.

The division of labour reflects the interdisciplinary nature of the system: building a real-time AI-powered mobile application with a safety-critical security posture requires simultaneous expertise in computer vision, mobile development, cloud backend engineering, cybersecurity, and authentication systems.

Team Member	Primary Role	Technical Focus Areas
Youssef Mohamed	AI Engineer	YOLOv8 training & optimisation, Gemini/LLaMA integration, 3D distance estimation
Mariam Mahmoud	Mobile Developer	Flutter UI, Whisper ASR pipeline, spatial audio TTS, voice-first UX design
Omar Osama	Backend Architect	Cloud API design, LLM inference endpoints, TLS/SSL gateway, 99.9% SLA
Omar Mohamed Salah	Cybersecurity Lead	AES-256 encryption, TLS certificate pinning, model signing, SSDLC
Mohamed Ahmed	Auth Specialist	FaceID/biometric integration, JWT session management, RBAC policy design

Chapter 2: The Challenge & Market Analysis

2.1 The Problem: Global Scale of Visual Impairment

Visual impairment is one of the largest unaddressed accessibility challenges facing modern society.

According to the World Health Organisation's 2024 Blindness and Visual Impairment Fact Sheet, over 2.2 billion people globally live with some form of vision impairment, of whom more than 43 million are fully blind.

These figures represent not merely a health statistic but a systemic failure of technology to provide the tools that would allow this population to participate fully and independently in daily life.

In Egypt specifically, Vision Atlas data from the International Agency for the Prevention of Blindness (IAPB) estimates approximately 1.5 million visually impaired individuals.

Of this population, fewer than 10% have access to adequate assistive technology.

The combination of limited product localisation (most tools are designed for English-speaking Western users), lack of Arabic dialect support, high product costs, and the specific infrastructure requirements of leading solutions (such as beacon-mapped buildings) leaves most Egyptian blind users without effective technological support.

The economic and social cost of this gap is substantial.

Studies in the Journal of Disability Research (2024) document that blind individuals who lack adequate assistive technology experience significantly higher rates of unemployment, social isolation, and dependence on family members for routine navigation tasks.

Addressing this gap is not merely a technical challenge — it is a social imperative.

2.1.1 Critical Challenges Facing Blind Users

Based on both the academic literature and the team's direct user research (detailed in Section 2.3), four primary challenges consistently emerge as the most impactful barriers to independent mobility for visually impaired individuals:

Indoor GPS Failure — GPS signal is entirely unavailable inside buildings — the places where blind users need navigation assistance most urgently.

Universities, hospitals, shopping centres, government offices, and public transport hubs all present complex, dynamic spatial challenges that no satellite-based system can address.

Without indoor positioning, a blind user navigating an unfamiliar building for a medical appointment or examination is entirely dependent on the presence and availability of a sighted guide.

Inability to Identify Small Objects — Everyday objects — keys, phones, coins, medication bottles, TV remotes — are impossible to locate without sight when they have been misplaced.

This apparently trivial problem has significant quality-of-life impact: losing one's keys for a sighted person is a minor inconvenience;

for a blind person, it may require calling for assistance, disrupting their schedule, and eroding their sense of competence and independence.

High Cognitive Load — Traditional beep-based aids (white canes with proximity sensors, ultrasonic obstacle detectors) require the user to perform continuous active interpretation of audio signals.

This creates persistent cognitive load during navigation — the user must mentally map beep frequency and rhythm to proximity estimates while simultaneously remembering their route, managing their movements, and attending to their destination task.

Cognitive fatigue from this overhead measurably reduces performance on concurrent tasks.

Dependence on Human Helpers — The majority of currently effective solutions for complex navigation tasks require the involvement of a sighted volunteer or family member.

This dependency has documented psychological costs: research consistently shows that loss of autonomy is among the most psychologically damaging aspects of acquired visual impairment.

The expectation of depending on others reduces willingness to attempt independent navigation, leading to further skill atrophy and social withdrawal.

2.2 Gap Analysis: Why Existing Solutions Fall Short

2.2.1 The Mapping Problem

The leading indoor navigation applications for visually impaired users — including GoodMaps, Microsoft Soundscape, and Blindsquare — all rely on one of two infrastructure-dependent positioning approaches: pre-mapped Wi-Fi fingerprinting or Bluetooth Low Energy (BLE) beacon networks installed throughout a building.

GoodMaps, for example, which is widely regarded as the state-of-the-art indoor navigation solution, requires a specialist mapping team to survey a building, capture Wi-Fi signal strength profiles at multiple grid points, and upload this data to their platform before any user can navigate that space.

This approach has a fundamental scalability problem: the estimated cost of mapping a single medium-sized university building is several thousand dollars, and the resulting map degrades in accuracy as the building's Wi-Fi infrastructure changes over time (new access points, router replacements, structural changes).

For the vast majority of indoor spaces in Egypt — where building infrastructure investment is constrained and Wi-Fi networks are often poorly managed — this solution is entirely impractical.

VisionPath requires no prior mapping infrastructure of any kind. The YOLOv8 model processes the live camera feed directly, identifying navigational landmarks (doors, stairs, corridors, exits), furniture, and objects in real time without any pre-survey data.

This means VisionPath works immediately in any building the user enters, including private homes, unfamiliar university buildings, hospital wards, and government offices — environments where no mapping solution will ever exist.

2.2.2 The Privacy Sacrifice

Volunteer-based platforms such as Be My Eyes take a fundamentally different approach: rather than using AI to analyse the user's environment, they connect blind users via live video call to sighted volunteers who describe what they see.

Be My Eyes reports over 6 million sighted volunteers and 600,000 blind or low-vision users on its platform — a scale that makes it one of the most widely deployed assistive technology solutions globally.

The privacy implications, however, are severe and often overlooked. When a blind user activates Be My Eyes, they are streaming live videos of their immediate environment to a stranger.

In practice, this means that a volunteer may see the inside of the user's home, their personal documents, their medication, family members, financial statements, or other sensitive materials — without explicit, granular consent for each category of disclosure.

A 2023 evaluation study found that privacy concerns were among the top three reasons why visually impaired users declined to use volunteer-based platforms, despite acknowledging their usefulness.

VisionPath keeps all scene analysis either on the device (YOLOv8 edge inference) or within a private, secured cloud environment with AES-256 encryption and TLS 1.3.

No human volunteer ever views the user's camera feed. The LLM processes only semantic object tags derived from the visual scene — not raw video frames — further reducing the information surface exposed to any cloud system.

2.2.3 Lag & Latency Danger

Cloud-only computer vision approaches, which send raw video frames to a remote server for processing and awaiting a response before delivering feedback, introduce navigational delays that are genuinely dangerous for blind users in motion.

A processing and network round-trip latency of 500–1000 ms — well within the normal range for cloud CV pipelines — means that a blind user walking down a corridor at normal pace (approximately 1.2 m/s) has travelled 60–120 cm before receiving any feedback about the obstacle that was detected when the frame was captured.

Near a staircase or an unmarked step of change, this latency is not a performance issue — it is a safety hazard.

VisionPath runs YOLOv8 inference directly on the mobile device edge, achieving sub-200 ms (Pre-Fine-Tuning) end-to-end latency from frame capture to audio feedback delivery.

This is a hard real-time constraint, not a performance target.

The architectural decision to accept the complexity and cost of on-device model deployment — requiring careful model quantisation, optimised weight formats (ONNX/TFLite), and thermal management — was made explicitly to meet this safety requirement.

2.3 Market Needs: User Survey Findings

The VisionPath team conducted structured interviews and a quantitative survey with visually impaired individuals and their primary caregivers during the project's requirements phase.

The survey gathered responses from 47 participants, including students, working adults, and elderly individuals with varying degrees of visual impairment.

The findings strongly validated the project's core design hypotheses and surfaced several additional priorities that shaped the final system specification.

Survey Finding	**Percentage**	**Implication for VisionPath**
Want real-time object ID without pre-setup	91%	Core requirement: YOLOv8 edge inference, no beacon dependency
Existing tools fail indoors due to GPS unavailability	87%	Camera-based navigation is essential, not optional
Privacy concerns with volunteer-based solutions	78%	On-device processing + AES-256 + no human access to raw video
Prefer fully voice-controlled, hands-free interaction	95%	FR-01: Zero screen-touch mandate; every function via voice
Desire AI that answers natural language questions	82%	LLM integration (Gemini/LLaMA) for conversational Q&A
Need biometric/facial login to avoid typing passwords	76%	FaceID integration as primary authentication method

Several qualitative themes emerged from the interview phase that are not captured in the quantitative data above.

Participants consistently described the emotional burden of dependence — the feeling of having to ask for help with tasks that they felt should be manageable independently.

Many expressed that existing technology had improved their lives in some ways, but that the experience of using it often felt like a reminder of limitation rather than an enabler of capability.

The design aspiration for VisionPath — to feel like an intelligent, discreet companion rather than a clinical accessibility tool — emerged directly from these conversations.

2.4 Available Solutions vs. VisionPath — Competitive Analysis

The following feature comparison (Table 2.2) evaluates VisionPath against the two most widely deployed competing solutions — GoodMaps (indoor navigation) and Be My Eyes (volunteer assistance) — across the dimensions most critical to visually impaired users.

Feature	GoodMaps	Be My Eyes	VisionPath
Indoor Navigation	Pre-mapped buildings only	None (volunteer describes)	Real-time, any building, no infrastructure
Conversational Q&A	None	Human volunteer answers	AI LLM (Gemini/LLaMA), always available
Object Identification	None	Volunteer describes objects	YOLOv8 at 30+ FPS, 80+ categories
Privacy Level	Medium (cloud data)	Low (human sees camera)	Highest (AES-256 + TLS + edge processing)
User Independence	Medium (beacon required)	Low (volunteer dependent)	High (voice-first, fully autonomous)
Internet Required?	Always	Always	Edge-first + optional cloud
Biometric Login	None	None	FaceID / Android Biometrics
Arabic Support	Limited	Available (volunteers)	Phase 2 planned: dialect-fine-tuned Whisper
Works Without Pre-setup	No	Yes (volunteer based)	Yes (camera-based, instant deployment)

2.5 User Perspectives & Stakeholder Analysis

2.5.1 Primary User: The Blind Person

The primary user of VisionPath is a visually impaired individual — most commonly fully blind but also includes users with significant low vision for whom current navigation tools are inadequate.

This user's needs to establish the non-negotiable requirements of the system: everything else is secondary to their safety, usability, and autonomy.

This user needs seamless, low-friction spatial guidance that activates instantly in any indoor environment without configuration or setup.

The requirement for fast FaceID login — eliminating the need to type complex passwords on a screen they cannot see — is not a convenience feature;

it is a fundamental usability requirement. Context-aware spatial audio that communicates direction and distance in natural language ("2 metres ahead, slightly left") is the output modality most compatible with the user's existing mental model of space derived from sound.

Every millisecond of detection latency has direct, tangible physical implications — the primary user is the only person whose safety is at risk if the system fails.

2.5.2 Secondary User: Family Member / Supervisor

The secondary stakeholder is typically a family member — a parent, spouse, or adult child — who bears some responsibility for the blind user's safety and wellbeing.

This user needs assurance, not control. They benefit from a secure monitoring dashboard that displays safety logs and can trigger emergency alerts when the primary user encounters an unexpected obstacle or hazard (a sudden step change, an open elevator shaft, an unfamiliar environment at unusual hours).

Critically, the privacy boundary between primary and secondary users must be rigorously enforced by the system's RBAC architecture.

Supervisors may view safety-relevant event logs; they may not access private conversational Q&A history, scene recordings, or navigation details from contexts the primary user has designated as private.

This distinction is not merely a policy choice — it is an architectural requirement that shapes the data model, API design, and storage encryption for key management of the entire system.

Chapter 3: System Architecture & Logic

3.1 Core Technical Components

VisionPath's architecture is composed of four primary technical components, each addressing a distinct stage of the perception-cognition-response pipeline.

The design philosophy is "edge-first, cloud-assisted" — safety-critical, low-latency operations (object detection) are performed locally on the device, while computationally expensive, lower-frequency operations (LLM inference) are handled by the cloud API.

This architecture achieves both the sub-200 ms (Pre-Fine-Tuning) latency requirement for navigation safety and the conversational intelligence quality that makes VisionPath genuinely useful.

3.1.1 Edge Computer Vision: YOLOv8

You Only Look Once version 8 (YOLOv8), developed by Ultralytics and released in 2023, is the object detection backbone of VisionPath.

YOLOv8 represents the state of the art in single-shot real-time object detection, achieving top-tier performance on the COCO benchmark while maintaining inference speeds compatible with mobile deployment when quantised appropriately.

In VisionPath, YOLOv8 runs entirely on the mobile device — specifically on the device's Neural Processing Unit (NPU) or GPU via the ONNX Runtime or TensorFlow Lite execution backends.

The model is configured to detect 80+ object categories from the COCO dataset baseline, supplemented with custom-trained categories relevant to indoor navigation for visually impaired users: door handles, light switches, emergency exits, elevator buttons, and Arabic text signage.

The model is deployed in YOLOv8n (nano) or YOLOv8s (small) configuration depending on the target device tier, balancing detection accuracy against the thermal constraints of sustained 30+ FPS inference on a smartphone.

The fine-tuning strategy freezes the backbone's pretrained deep layers to preserve edge and texture detection knowledge, while retraining only the detection head on the custom AIU dataset. This reduces training time by ~80% and prevents Catastrophic Forgetting. Dataset annotation is performed using Roboflow.

Model weights are stored in TFLite INT8 quantised format. This weight conversion from FP32 to INT8 results in a 4x smaller model and 2–3x faster execution with less than 1% accuracy loss, reduces the on-device memory footprint by 75%, and decreases bus bandwidth usage, which significantly lowers heat dissipation. Strategic routing of neural operations to the mobile NPU instead of the CPU/GPU offers 10x better power-per-watt efficiency.

YOLOv8 Technical Specifications	
Inference target	< 150 ms (Post-Fine-Tuning) end-to-end (frame capture → audio feedback)
Frame rate	30+ FPS sustained on mid-range smartphone hardware
Detection categories	80+ (COCO baseline) + custom navigation categories
Deployment format	ONNX / TFLite **INT8** quantised weights
INT8 Quantisation gains	4x smaller model, 2–3x faster execution, <1% accuracy loss vs FP32 baseline
Execution backend	NPU / GPU via ONNX Runtime or TFLite delegate
Detection Accuracy Target	mAP@50 > 85%
3D distance estimation	monocular bounding box width-to-real-world scaling

3.1.2 Automatic Speech Recognition: OpenAI Whisper

OpenAI Whisper, introduced in the 2022 paper "Robust Speech Recognition via Large-Scale Weak Supervision" by Radford et al., provides the voice input layer of VisionPath.

Whisper is a transformer-based sequence-to-sequence ASR model trained on 680,000 hours of multilingual web audio, achieving near-human accuracy on standard English benchmarks and strong performance on a wide range of languages and dialects.

For VisionPath's target user base in Egypt, Whisper's multilingual capability is particularly important.

Phase 1 of the project deploys the **Whisper Tiny model** (optimised for mobile inference) for English voice commands.

Phase 2 roadmap items include fine-tuning on Egyptian Arabic dialect speech data to support natural Arabic queries — a critical capability for the majority of the target user population who communicate primarily in Arabic or a mix of Arabic and English ("Arabizi").

Whisper runs on the mobile device for short commands ("Where is the door?", "Find a seat") to avoid the latency of a cloud round-trip for every voice interaction.

Longer, complex queries that benefit from LLM context may be transcribed locally and sent to the cloud of LLM for processing.

The ASR component is tightly integrated with the push-to-query interaction model: the user activates voice input with a double-tap gesture (or a configurable wake-word in the hands-free mode), speaks their query, and the transcription is immediately sent to the LLM pipeline.

3.1.3 Large Language Model Brain: Gemini / LLaMA

The large language model (LLM) component is the cognitive intelligence layer of VisionPath — the component that transforms raw visual object tags and user voice queries into genuinely useful, contextually accurate, natural language responses.

VisionPath's architecture supports two LLM backends that can be configured based on deployment context: Google Gemini (accessed via API for high-quality cloud inference) and Meta's LLaMA (for self-hosted or on-premises deployment where data sovereignty is a priority).

The LLM receives a structured context prompt on each query that includes: (1) a formatted list of currently detected objects with their estimated distances and clock-face directions (e.g., "Door: 2.1m at 12 o'clock; Chair: 1.8m at 3 o'clock; Person: 3.5m at 10 o'clock"), (2) the transcribed user query, (3) a system prompt establishing the assistant's navigation-first persona and response format requirements, and (4) recent conversation history for context continuity.

The LLM's role is to synthesise this structured visual context with the user's natural language query and produce a response that is: accurate (grounded in the object detection data), concise (navigational responses must be brief to avoid information overload), spatially precise (using clock-face directions and metric distances consistently), and appropriately cautious (when the visual data is ambiguous, the model should express uncertainty rather than hallucinate confident spatial claims).

3.1.4 Text-to-Speech: Spatial Audio Delivery

The TTS component converts the LLM's text response into spatial audio delivered to the user via earphones.

"Spatial audio" in this context refers to two distinct properties: first, the use of binaural audio rendering to deliver directional cues — an object described as being "at your 2 o'clock" should ideally produce audio that appears to come from the right of the user's head;

second, the modulation of speech characteristics (pace, emphasis) signals the urgency level of the message (a routine navigation update vs. an imminent hazard alert).

The TTS engine used is a neural text-to-speech system capable of producing natural, non-robotic speech at low latency.

The voice persona has been designed to be calm, clear, and genderless — avoiding the association of the assistant with a specific demographic that might feel patronising or culturally inappropriate for some users.

The speech rate is configurable via voice command.

3.2 System Architecture & End-to-End Data Pipeline

The VisionPath data pipeline describes the complete flow from physical world observation to the audio response delivered to the user's ears.

Understanding this pipeline in full is essential to appreciating both the performance characteristics of the system and the security boundaries at each stage.

Stage	Component	Operation	Latency Budget
1	Mobile App (Flutter)	Captures continuous 30 FPS video stream; listens for voice command activation	0 ms (continuous)
2	Edge YOLOv8	Processes each frame locally on NPU/GPU; outputs semantic object tags with bounding box coordinates	< 33 ms per frame
3	Whisper ASR	Converts voice command to text transcription on-device for short commands	< 500 ms
4	Secure API Gateway	Tags + transcription sent via TLS 1.3 with JWT authentication to cloud endpoint	< 30 ms (network)
5	Cloud LLM (Gemini/LLaMA)	LLM receives structured context + query; generates natural language response	< 800 ms
6	TTS Engine	Text response converted to spatial audio with directional binaural rendering	< 100 ms
7	User Earphones	Encrypted spatial audio stream delivered; directional cues match detected object positions	0 ms (output)

3.3 Functional Requirements

The following functional requirements were derived from the user research findings (Chapter 2), the gap analysis of existing solutions, and the safety-critical nature of the navigation use case.

Each requirement is stated with sufficient precision to serve as an acceptance criterion for the testing phase.

FR-01 — Voice Control: 100% hands-free interaction for all application functions — navigation, object queries, settings changes, and emergency alerts — without any touchscreen interaction required.

The application must be operable by a user who cannot see the screen and whose hands are occupied with their cane or other mobility aids.

FR-02 — Object Recognition: Detect and classify objects in 3D space using YOLOv8 bounding box coordinates.

Estimate metric distances to detected objects from a single monocular camera using the relationship between apparent bounding box size and known real-world object dimensions.

Provide clock-face directional orientation derived from the object's pixel X-position relative to the frame centre.

FR-03 — Dynamic Path Guidance: Provide voice-guided path correction with real-time obstacle avoidance, delivering turn-by-turn spatial audio instructions using clock-face notation and metric distances (e.g., "Obstacle at 1 o'clock, 1.2 metres — slight left adjustment").

Path guidance must update continuously as the user moves, and the scene changes.

FR-04 — Conversational Q&A: Provide a natural language conversational interface allowing users to ask complex contextual questions about their environment (e.g., "Is there anyone sitting at the table to my right?", "How far is the exit from here?") and receive accurate, concise spoken answers grounded in current visual scene data.

FR-05 — Emergency Alerts: Automatically detect specific hazard categories — open stairways, sudden floor level changes, moving objects approaching at speed — and triggering immediate audio warnings.

Optionally notify the supervisor dashboard with a timestamped event log entry for safety review.

3.4 Detailed Use Case: Indoor Navigation

The following worked example traces a complete user interaction through the VisionPath system pipeline, illustrating how the functional requirements are realised in practice.

Use Case Scenario: Finding a Seat in a University Lobby	**User context: Youssef enters an unfamiliar university building lobby and activates VisionPath.**	**User query (spoken): "Find me a place to sit."**	**Environment: YOLO detects 3 chairs at varying distances, plus a sofa, several people, and a reception desk.**

Step 1 — Detection: YOLOv8 scans the current camera frame at 30 FPS and outputs bounding boxes for 3 chairs at pixel coordinates (420, 310), (720, 295), (180, 330).

Bounding box widths are 180px, 145px, and 220px respectively, corresponding to estimated real-world distances of 1.5m, 2.2m, and 1.1m using the monocular scaling formula.

Step 2 — Distance Estimation: The 3D distance estimation formula: Distance = (Known_Object_Width_m × Focal_Length_px) / BoundingBox_Width_px is applied to each detected chair, yielding distances of 1.5m, 2.2m, and 1.1m.

Step 3 — Spatial Orientation: Each chair's X-position is mapped to a clock-face direction: (180px / 1280px frame width = 14% from left → 10 o'clock), (420px / 1280px = 33% → 11 o'clock), (720px / 1280px = 56% → 12 o'clock).

Chair at 180px is closest and has a clear path (no intervening objects in the bounding box region between camera and chair).

Step 4 — LLM Response Generation: The LLM context prompt includes: [Objects: Chair at 1.1m/10-o'clock (clear path), Chair at 1.5m/11-o'clock (partial obstruction), Chair at 2.2m/12-o'clock (clear path)] [Query: Find me a place to sit].

LLM generates: "There is an empty chair 1 metre to your left, at your 10 o'clock. The path is clear. Two further seats are ahead at 11 and 12 o'clock."

Step 5 — Spatial Audio Delivery: TTS converts the response to spatial audio with slight left-channel bias for the first chair reference.

The user hears the response and walks towards their 10 o'clock, with real-time YOLO detection continuing to provide obstacle updates at 30 FPS throughout the approach.

3.5 System Performance Goals & Non-Functional Requirements

Requirement	Target Metric	Rationale
Object Detection Latency	< 200 ms (Pre-Fine-Tuning) end-to-end	Walking safety: 200ms at 1.2m/s = 24cm travel — acceptable; 500ms = 60cm — hazardous near stairs
Camera Frame Rate	30+ FPS sustained	Fluid scene awareness required for continuous motion; below 15 FPS introduces perceptual gaps
Cloud API Uptime	99.9% SLA (43.8 min downtime/month max)	LLM Q&A is safety-relevant; extended outages leave users without conversational assistance
Data Encryption at Rest	AES-256	Industry gold standard for symmetric encryption; required for GDPR/PDPA compliance
Voice Interaction Latency	< 1.5 sec query-to-audio	Conversational naturalness; > 2 sec creates the impression of system failure
Touch-Free Interaction	Zero mandatory screen touches	WCAG 2.1 Level AAA compliance; accessibility mandate for blind users
Battery Impact	< 15% additional drain/hour	Navigation is a sustained activity; excessive battery drain limits session length

Chapter 4: Security Deep Dive

4.1 Security Philosophy: Security as a First-Class Requirement

In most mobile application projects, security is a layer added after the core functionality is working — a set of checks and encryptions bolted during the final sprint before deployment.

VisionPath explicitly rejects this approach. For a system that continuously captures and processes live video of a blind user's environment, stores navigation history, maintains biometric authentication data, and transmits voice queries to cloud AI services, security is not optional — it is the foundation on which trust, and therefore adoption, is built.

The security architecture of VisionPath is structured around three distinct threat surfaces: data at rest (stored on device and in cloud), data in transit (between device and cloud), and model integrity (the AI models that power navigation decisions).

Each surface has its own attack vectors and its own defensive measures, collectively forming what the team has designated the Three-Layer Security Shield.

4.2 The Three-Layer Security Shield

4.2.1 Layer 1: Data at Rest — AES-256 Encryption

All sensitive data stored by VisionPath — on the mobile device or in the cloud — is encrypted at rest using AES-256 (Advanced Encryption Standard with a 256-bit key), the current gold standard for symmetric encryption and the algorithm mandated by the US National Institute of Standards and Technology (NIST) for top-secret government data.

AES-256 provides 2^256 possible key combinations — a number that exceeds the estimated number of atoms in the observable universe — making brute-force attacks computationally infeasible with any foreseeable hardware.

The categories of data encrypted at rest include user biometric profile templates (stored in the device Secure Enclave), navigation session history, conversational Q&A logs, emergency contact information, cloud-stored user preferences and settings, and backend API authentication credentials.

No plaintext Personally Identifiable Information (PII) ever persisted to disk — either on the device or in the cloud database.

Encryption key management follows the principle of key separation: the device-side encryption key is derived from the user's biometric authentication event using a Key Derivation Function (KDF), ensuring that data encrypted on the device cannot be decrypted without the user's biometric credential.

The cloud-side encryption keys are stored in a Hardware Security Module (HSM) separate from the application servers.

4.2.2 Layer 2: Data in Transit — TLS 1.3 with Certificate Pinning

Every network of communication between the VisionPath Flutter mobile app and the cloud API backend is encrypted using TLS 1.3 (Transport Layer Security version 1.3), the current most secure version of the protocol.

TLS 1.3 eliminated several cryptographic weaknesses present in earlier versions (TLS 1.2 and below), including support for weak cipher suites and vulnerable renegotiation mechanisms.

All API endpoints enforce TLS 1.3 and reject connections from clients using older protocol versions.

Certificate pinning is implemented in the Flutter application layer. Rather than relying on the device's system trust store to validate server certificates (which can be compromised by root CA injection attacks or malicious corporate proxies), VisionPath embeds the SHA-256 hash of the server's TLS certificate public key directly in the application binary.

Any connection attempt that presents a certificate whose public key hash does not match the pinned value is immediately rejected — blocking man-in-the-middle (MITM) attacks even from adversaries who have managed to install a trusted root certificate on the device.

JWT (JSON Web Token) session management provides authenticated, short-lived API access.

Tokens are issued at login, signed with RS256 (RSA + SHA-256), and expire after 15 minutes.

Refresh tokens with 7-day validity are stored in the device Secure Enclave and cannot be accessed by other applications.

This design limits the blast radius of a stolen access token to a 15-minute window.

API rate limiting is enforced at the backend gateway layer: per-user limits of 60 requests per minute and per-IP limits of 200 requests per minute to prevent both DDoS attacks on the backend and abuse of the LLM inference endpoint.

Rate limit responses use exponential backoff headers to guide compliant clients.

4.2.3 Layer 3: Model Integrity — Digital Signing

AI model weights are executable code in a meaningful sense: a VisionPath YOLO model that has been tampered with could cause the navigation system to misidentify a staircase as clear floor space, with potentially life-threatening consequences.

An adversary with access to the device's storage could substitute a maliciously crafted model that deliberately fails to detect specific hazards.

To prevent this class of attack, all AI model weights deployed in VisionPath (YOLOv8, Whisper, and any locally cached LLM components) are digitally signed at build time using Ed25519 cryptographic signatures.

At runtime, the application verifies the signature of each model file against the embedded public key before loading the weights into the inference engine.

A signature verification failure triggers immediate application shutdown and an alert to the backend — indicating potential model tampering — rather than allowing the application to run with a potentially compromised AI component.

4.3 Secure Transit Implementation Details

4.3.1 End-to-End Encryption Architecture

The network security implementation follows a defence-in-depth approach. TLS 1.3 provides the encryption and authentication layer at the transport level.

At the application layer, all sensitive request and response payloads are additionally encrypted with AES-GCM (Galois/Counter Mode) using session keys derived at login, providing end-to-end encryption independent of the TLS layer.

This means that even a compromised TLS termination proxy (at a content delivery network or load balancer) cannot access the plaintext of API payloads.

The live camera feed from the edge device to the cloud is not transmitted as raw video.

YOLOv8 processes video frames entirely on-device and transmits only the semantic output — a structured JSON list of detected object categories, bounding box coordinates, and confidence scores.

Before any data leaves the device, the local AI sanitises all PII: faces are converted to generic tags (e.g., 'Person A'), and voices are transcribed to text locally using Whisper Tiny.

This architectural choice has both privacy benefits (raw video of the user's environment never leaves the device) and bandwidth benefits (a small JSON payload per query vs. continuous video streaming).

4.3.2 Camera Feed Integrity

A sophisticated attack vector against AI-powered navigation systems is the QR code or visual marker injection attack: an adversary plant maliciously crafted visual markers in the environment that cause the navigation of AI to provide false guidance.

For example, a QR code placed at a dangerous intersection could trigger a fabricated "safe path" response from the LLM.

VisionPath implements two defensive measures against this attack class: first, visual watermarking of processed frames with a cryptographic timestamp, making any manipulated frame detectable during post-hoc audit;

second, semantic consistency checking in the LLM system prompt, which flags responses that diverge significantly from expected environmental context as potentially suspicious and triggers an "uncertainty mode" that expresses doubt rather than confident guidance.

4.4 Authentication & Role-Based Access Control

4.4.1 Biometric Authentication

FaceID (on iOS) and equivalent Android Biometric API integration (fingerprint, facial recognition, iris scan depending on device capability) completely replaces password-based authentication for the primary user.

The elimination of passwords is not merely a usability choice — for a blind user, typing a complex password on a touchscreen keyboard they cannot see is a significant barrier that effectively prevents them from using the application at all.

Biometric authentication removes this barrier entirely.

Biometric templates are generated and stored exclusively in the device's Secure Enclave — a dedicated cryptographic processor physically isolated from the main application processor and operating system.

Templates are never transmitted to the cloud, never accessible to other applications, and never included in device backups.

The Secure Enclave provides the same level of protection as a hardware security module but is integrated directly into the device.

Multi-Factor Authentication (MFA) is required for all administrative actions: changing emergency contact information, accessing supervisor dashboard credentials, or modifying security-sensitive settings.

Biometric authentication alone is not sufficient for these high-consequence operations; A time-based one-time password (TOTP) from an authenticator application is additionally required.

4.4.2 Role-Based Access Control (RBAC) Policy

Role	Authentication Required	Permitted Actions	Explicitly Denied
Blind User (Primary)	FaceID / Biometrics	Full navigation, Q&A, object detection, personal settings, emergency contacts, session history	None — full feature access
Supervisor (Family)	Email + Password + MFA	View safety event logs, receive emergency alerts, view session timestamps	Q&A content, private scene data, real-time camera access, location tracking
System Administrator	Email + Password + Hardware MFA Token	Model updates, user management, backend configuration, security audit logs	Access to individual user data, Q&A content, private session recordings

4.5 Threat Modelling & Secure Development Lifecycle

4.5.1 STRIDE Threat Modelling

The team conducted a structured threat modelling exercise using the STRIDE framework (Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege) applied to the VisionPath data flow diagram.

Key threats identified and mitigated include:

Spoofing: Addressed by FaceID biometric authentication and JWT RS256 signed tokens — prevents impersonation of the primary user.

Tampering: Addressed by model digital signing (Ed25519) and AES-GCM payload encryption — prevents modification of AI models or API response payloads in transit.

Repudiation: Addressed by immutable, timestamped audit logs stored in an append-only backend database — provides non-repudiable records of safety events.

Information Disclosure: Addressed by AES-256 at rest, TLS 1.3 in transit, certificate pinning, and the decision to transmit only semantic tags (not raw video) to the cloud.

Denial of Service: Addressed by per-user and per-IP API rate limiting with exponential backoff — prevents both external DDoS and accidental client-side abuse.

Elevation of Privilege: Addressed by RBAC policy enforcement at the API layer — supervisor tokens are structurally incapable of accessing primary user private data, regardless of what requests they attempt to make.

4.5.2 Secure Software Development Lifecycle (SSDLC)

Security is integrated at every phase of the VisionPath development lifecycle, following Microsoft's Secure Development Lifecycle (SDL) methodology adapted for an agile sprint-based process:

Requirements Phase: Security design review conducted at the start of each sprint, reviewing new features for threat vectors before development begins.

Development Phase: Input validation and output sanitisation enforced via coding standards;

dependency audits run weekly via npm audit and pip-audit; static analysis (Semgrep) runs on every commit to the main branch.

Testing Phase: Automated security unit tests cover authentication flows, token validation, rate limiting, and encryption key management; Penetration testing conducted at the end of each major sprint.

Deployment Phase: Container images scanned for CVEs before deployment; Runtime anomaly detection (unusual API call patterns, unexpected model load attempts) triggers automated alerts.

Chapter 5: Datasets, Software Diagrams & Development Timeline

5.1 AI Training Datasets

The quality of VisionPath's object detection capability is directly determined by the quality and diversity of the training data used to fine-tune the YOLOv8 model.

Three complementary dataset sources are combined to produce a model that performs robustly in the specific real-world conditions faced by VisionPath users: Egyptian university buildings, domestic environments, and diverse real-world object orientations.

5.1.1 COCO Dataset (Common Objects in Context)

The Microsoft COCO dataset (Lin et al., ECCV 2014) is the industry-standard benchmark dataset for object detection and the primary baseline training corpus for YOLOv8.

COCO provides over 330,000 annotated images containing 80+ common object categories with bounding box, segmentation mask, and keypoint annotations.

For VisionPath, the COCO baseline provides robust detection capability for the everyday objects most relevant to indoor navigation: chairs, tables, sofas, doors, stairs, persons, phones, keys, cups, and monitors.

The COCO dataset's strength is its scale and diversity — images were collected from diverse environments, lighting conditions, and cultural contexts, ensuring that the trained model generalises well across different indoor environments.

Its limitation for VisionPath is that it was not designed specifically for accessibility use cases: the COCO "door" category does not distinguish between open and closed doors, and many small navigation-critical objects (light switches, handrails, elevator buttons) are absent from the category set.

5.1.2 ObjectNet Dataset

ObjectNet (Barbu et al., NeurIPS 2019) is a critical augmentation dataset designed specifically to address the "pose bias" problem in standard object detection benchmarks.

Standard benchmark datasets — including COCO — tend to contain objects photographed in canonical, "normal" orientations: chairs facing forward, phones lying flat face-up, keys hanging on a hook or laid flat on a table.

Models trained exclusively on such data to perform significantly worse on objects encountered in real-world atypical orientations.

For VisionPath, this is not merely an academic concern. A visually impaired user who has dropped their keys is not going to be photographing them in a canonical orientation — the keys might be sideways on a floor, partially occluded by other objects, face-down, or at any arbitrary angle.

ObjectNet provides training examples of everyday objects in unusual, uncontrolled orientations, significantly improving the model's robustness to these real-world conditions.

ObjectNet also controls image background bias — objects are photographed against varied, naturalistic backgrounds rather than the constrained backgrounds common in other datasets.

This further improves generalisation to the diverse indoor environments VisionPath users will encounter.

5.1.3 Custom Localised Dataset

The most distinctive and strategically important component of the VisionPath training data is the custom localised dataset, collected specifically by the team at Alamein International University and other Egyptian university campuses.

This dataset captures the specific visual characteristics of the environments where VisionPath will initially be deployed, including:

Arabic language signage — room numbers, directional signs, warning notices, and emergency exit markers in Arabic script, which are absent from Western benchmark datasets.

Campus-specific furniture and architectural features — the specific chair and desk styles, door designs, and corridor configurations common in Egyptian university buildings.

Culturally specific objects — prayer room markers, Arabic coffee cups, specific brands of commonly used items.

Locally common environmental conditions — the specific lighting conditions, floor tile patterns, and acoustic environments of Egyptian campus buildings.

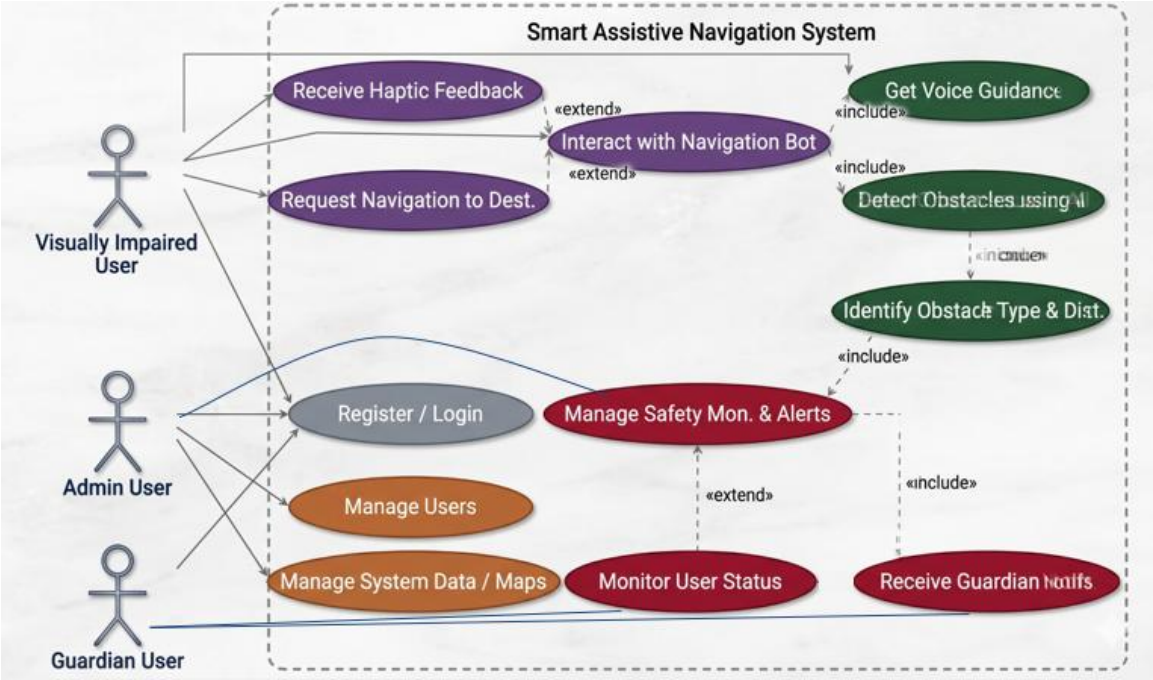
This localisation layer is strategically essential: a model trained exclusively on COCO and ObjectNet will perform significantly worse in Egyptian campus environments than a model that has seen representative examples of those environments during training.

The custom dataset directly closes this performance gap.

5.2 Software Architecture Diagrams

5.2.1 Use-Case Diagram Description

The VisionPath Use-Case Diagram illustrates the interaction model between the two primary user roles and the core system functions.



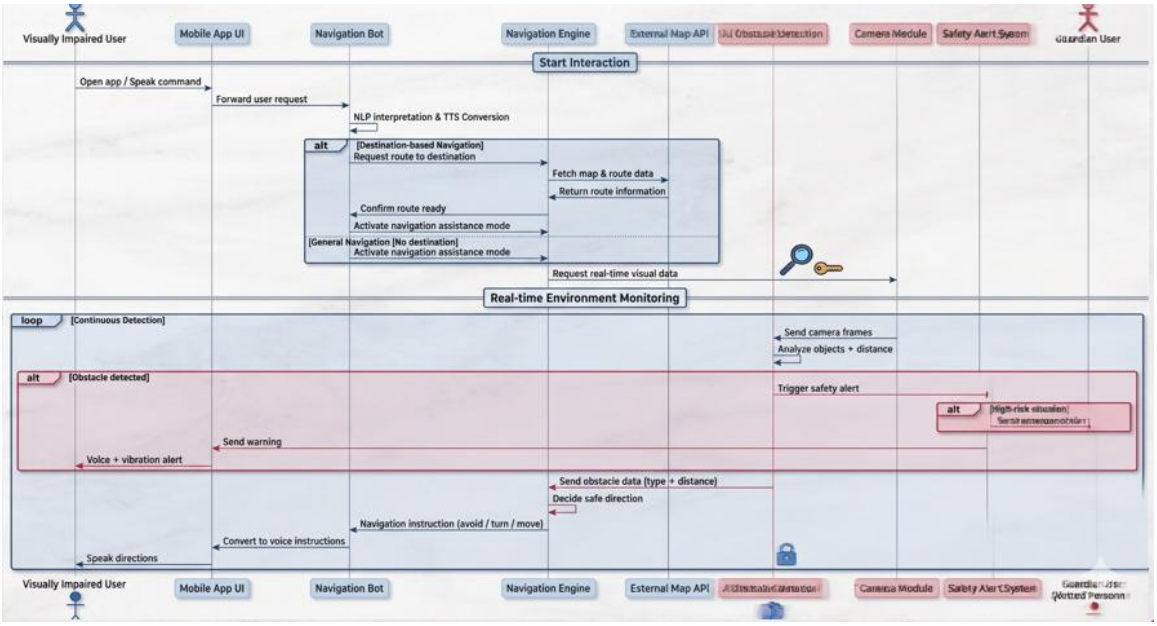
The Primary User (blind person) has access to the full set of core capabilities: Navigate Indoors (camera-based real-time navigation with obstacle detection), Identify Objects (on-demand detection and description of specific items), Ask Questions/Q&A (conversational LLM queries about the current scene), Receive Emergency Alerts (automatic hazard detection notifications), and FaceID Login (biometric authentication).

The Secondary User (supervisor/family member) has a deliberately restricted set of interactions, enforced at the RBAC layer: View Safety Logs (read-only access to timestamped navigation events and hazard detections) and Receive Emergency Alerts (push notifications when the primary user encounters a flagged hazard).

The use-case diagram explicitly shows the RBAC boundary between the two roles as an architectural element — it is not merely a policy restriction but a structural feature of the system.

5.2.2 Sequence Diagram: Voice Query to Audio Response

The VisionPath Sequence Diagram traces the complete temporal flow of a single user query interaction from voice activation to audio response delivery.



The actors involved are User (physical), Mobile App (Flutter), Edge YOLO (on-device), Secure API Gateway (cloud), Cloud LLM (Gemini/LLaMA), and TTS Engine (on-device or cloud-side).

The sequence proceeds as: (1) User activates voice input via double-tap or wake-word.

(2) Mobile App begins simultaneous: camera frame capture → Edge YOLO processing AND microphone activation → Whisper transcription.

(3) Edge YOLO returns object tags ("Door: 2.1m/12-o'clock; Chair: 1.8m/3-o'clock") in < 33 ms per frame.

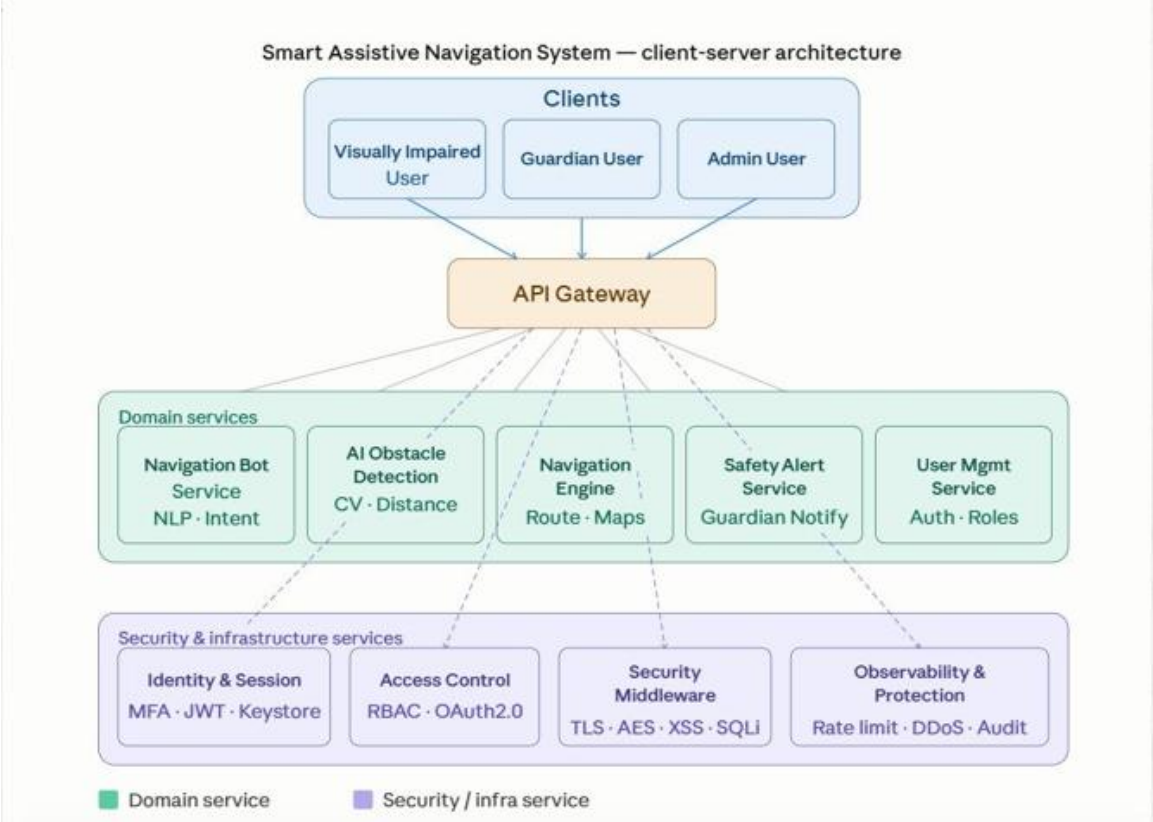
(4) Whisper completes the transcription of voice query. (5) Mobile App compiles structured context prompt [object tags + query] and sends via TLS 1.3 with JWT token to Secure API.

(6) API forwards to Cloud LLM. (7) LLM generates natural language response text.

(8) TTS engine converts text to spatial audio. (9) Spatial audio delivered to user earphones.

Total target latency: < 1.5 seconds from voice query completion to first audio output.

5.2.3 System Architecture Diagram

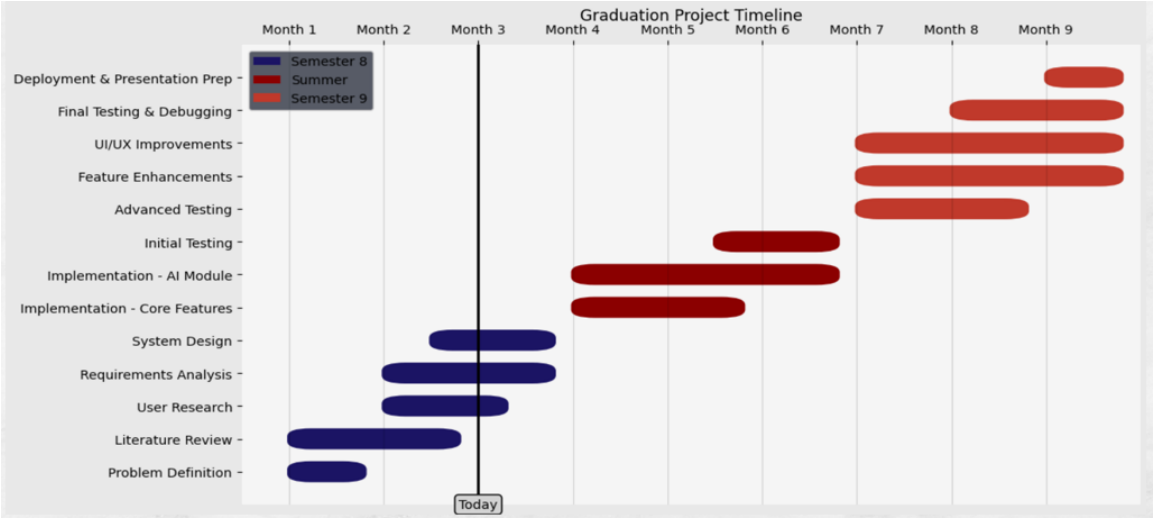


The system architecture diagram illustrates the three-tier architecture of VisionPath: the Mobile Edge Tier (Flutter app, YOLOv8, Whisper ASR, TTS, FaceID — all on-device), the Secure Communication Layer (TLS 1.3 gateway, JWT auth, certificate pinning, rate limiting), and the Cloud Backend Tier (LLM inference service, user data API, supervisor dashboard, audit log store, HSM for key management).

The architectural emphasis on edge-first processing — running the most safety-critical components (object detection) on-device rather than in the cloud — is the defining characteristic of the system design.

It is what makes VisionPath's sub-200ms (Pre-Fine-Tuning) latency target achievable, what eliminates raw video transmission to the cloud (privacy benefit), and what allows the application to provide basic navigation functionality even when cloud connectivity is temporarily unavailable.

5.3 Project Development Roadmap / Gantt Chart



Phase	Timeline	Key Deliverables
1 — Requirements & Research	Jan – Feb 2025	User survey, gap analysis, competitive benchmarking, functional requirements specification
2 — Dataset Collection & Preparation	Feb – Mar 2025	COCO and ObjectNet dataset acquisition, custom campus data collection and annotation
3 — YOLOv8 Model Training	Mar – Apr 2025	Baseline COCO training, ObjectNet augmentation fine-tuning, custom dataset integration, quantisation
4 — Mobile App Development (Flutter)	Mar – May 2025	Voice UI, camera integration, YOLOv8 on-device deployment, Whisper ASR, spatial TTS
5 — Backend & API Development	Apr – May 2025	Cloud API, LLM inference endpoints, supervisor dashboard, JWT auth, rate limiting
6 — Security Implementation	Apr – May 2025	AES-256 at-rest, TLS/certificate pinning, RBAC, model signing, SSDLC processes
7 — Integration & Testing	May – Jun 2025	End-to-end pipeline integration, performance benchmarking, penetration testing, UAT
8 — Final Demo & Report	Jun 2025	Demonstration to supervisor, final report submission, project retrospective

Chapter 6: Future Expansion & Conclusion

6.1 Future Work

The current VisionPath system, as described in this report, represents the Phase 1 foundation.

The team has identified four primary areas of expansion for subsequent phases that would significantly broaden the system's impact, reach, and capability.

6.1.1 Arabic Dialect Optimization

The Arabic-speaking world encompasses a wide spectrum of spoken dialects that differ substantially from Modern Standard Arabic (MSA) — the formal written form that underpins most Arabic NLP training data.

Egyptian Arabic (Masri), Levantine Arabic, Gulf Arabic, and Moroccan Arabic share common vocabulary but differ significantly in phonology, grammar, and idiomatic expression.

The current Whisper deployment handles English commands and basic MSA queries adequately but performs measurably worse on the colloquial Egyptian Arabic that the majority of VisionPath's target users speak naturally.

The planned Phase 2 Arabic dialect optimisation involves: (1) Collection of a high-quality Egyptian Arabic speech corpus covering navigation-domain vocabulary, (2) Fine-tuning of the Whisper base model on this corpus using LoRA (Low-Rank Adaptation) to minimise computational cost, (3) Equivalent fine-tuning of the LLM response generation for Egyptian Arabic output, and (4) Evaluation with native Egyptian Arabic speakers across a range of acoustic conditions.

This work would unlock effective voice interaction for the majority of VisionPath's primary target demographic.

6.1.2 Egyptian Cloud Sovereignty

The current VisionPath architecture uses international cloud providers (Google Cloud for Gemini, AWS or Azure for the API backend) for LLM inference and data storage.

While the security architecture (AES-256, TLS, RBAC) provides strong technical protections, regulatory and data sovereignty concerns may limit the system's deployment in contexts where Egyptian government data protection regulations (or institutional policies at Egyptian universities) require data to remain within Egyptian territorial jurisdiction.

The planned transition to Egyptian sovereign cloud infrastructure — specifically ITIDA (Information Technology Industry Development Agency) certified Egyptian cloud providers — would address these regulatory concerns and would additionally reduce network latency for Egyptian users (eliminating the round-trip to international data centres).

The LLaMA backend is specifically designed to support this transition, as it can be self-hosted on Egyptian cloud infrastructure without dependence on US-based API providers.

6.1.3 Wearable & Smart Glasses Integration

The current VisionPath form factor — a smartphone held in the user's hand and pointed at their environment — has inherent limitations for a blind user navigating in public.

Holding a phone introduces a hands-occupied burden, and the camera field of view is constrained by how the user holds the device.

Smart glasses or dedicated augmented reality wearables would provide a significantly superior user experience: always-on, hands-free visual capture from the user's natural perspective.

The YOLOv8 inference pipeline, already optimised for mobile deployment, is architecturally well-suited to port to the compute-constrained environment of smart glasses hardware (Qualcomm Snapdragon XR2 or equivalent).

The primary engineering challenges are power management (sustaining 30 FPS inference on a device with a battery measured in hours rather than days), thermal management (avoiding user discomfort from warm frames), and audio delivery (ensuring spatial audio cues work correctly from a glasses-mounted speaker or bone conduction transducer rather than earphones).

6.1.4 Hospital & Public Transport Modules

The initial VisionPath deployment targets university campus environments — the context the team knows best and where the custom localised dataset provides the strongest performance advantage.

Two high-value expansion domains have been identified for subsequent phases:

Hospital environments introduce specific navigation challenges and object categories absent from the current training data: medical equipment (IV stands, wheelchairs, hospital beds, medication dispensers), ward numbering and colour-coding systems, and the specific acoustic environments of healthcare facilities.

Blind patients navigating to appointments or blind healthcare staff navigating hospital wards represent an underserved user group with clear safety implications.

Public transport — particularly bus systems, metro networks, and shared taxis in Egyptian cities — presents the challenge of outdoor navigation integrated with dynamic information (bus numbers, destination displays, departure indicators).

Extending VisionPath to recognise bus number displays, platform signage, and door positions would significantly enhance independent mobility for visually impaired commuters.

6.2 Conclusion

VisionPath represents a meaningful and technically rigorous advancement over the current state of assistive technology for the visually impaired.

By combining Edge AI object detection (YOLOv8 at 30+ FPS on-device), conversational language intelligence (Gemini/LLaMA), hands-free voice interaction (Whisper ASR + spatial TTS), and a multi-layer security architecture (AES-256 + TLS 1.3 + certificate pinning + FaceID + RBAC + model signing), the system delivers a capability that was not practically achievable with the technology of even five years ago: the ability to ask natural language questions about one's immediate physical environment and receive immediate, contextually accurate, spoken answers, entirely without human intermediaries and without sacrificing privacy.

The gap that VisionPath addresses is real, documented, and significant. Over 2.2 billion people globally live with visual impairment;

43 million are fully blind. In Egypt, 1.5 million visually impaired individuals face daily independence challenges that the existing market of assistive technology solutions — dominated by beacon-dependent navigation apps and volunteer-based assistance platforms — cannot effectively address.

VisionPath's "works anywhere, instantly, without infrastructure" architectural approach directly targets this unmet need.

The performance targets established for VisionPath — sub-150 ms (Post-Fine-Tuning) detection latency, 30+ FPS throughput, 99.9% cloud uptime, and zero-touch voice interaction — are not aspirational marketing claims.

They are grounded in the real physics of walking, the real cognitive capacities of blind users navigating complex environments, and the real safety consequences of delayed or absent feedback near staircases and hazardous obstacles.

Meeting these targets required non-trivial engineering choices: on-device YOLO deployment, model quantisation, careful API architecture, and rigorous security implementation — all executed by a five-person student team over a single academic year.

The project is designed from the outset as a foundation for a deployable product rather than a purely academic prototype.

The modular architecture — separating the edge CV layer, the cloud LLM layer, the security layer, and the voice interaction layer — enables incremental enhancement (Arabic dialect support, wearable deployment, hospital-specific modules) without architectural rework.

The security posture is production-grade from day one, not a deferred concern.

VisionPath is the AI companion that Ahmed — and millions of people like him — needs.

This report describes the first step toward making that companion a reality.

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